



DEALER BEST PRACTICE SERIES

Ripper Guard Remove & Install Tool

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Ripper Guard Remove & Install Tool0

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1.0 Introduction

This Best Practice provides instruction for an easier way to replace ripper shank guards through the use of a purpose built lifting jig.

Changing a ripper shank protector is currently a difficult task. The single shank protector (9N-4621) on the D11T weighs 74kg and requires multiple technicians to maneuver the heavy guard onto the ripper tooth before it is positioned in order to install the retaining pins.

This maneuvering is often performed in an awkward position due to the ripper carriage overhead. This can result in unsafe lifting practices and potential injuries.

Use of this tool reduces the need for multiple technicians required to complete this task while removing many of the risks associated with ripper protection replacement methods.

2.0 Best Practice Description

Tool Design:

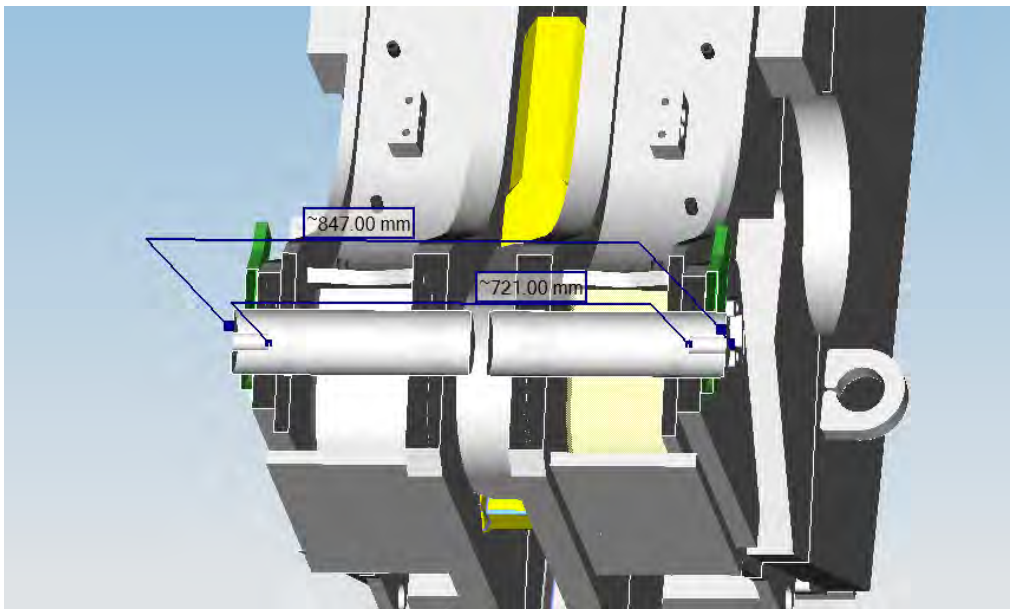
Tool is required to be installed in the lower ripper pin mounts.

Key dimensions for D11R/D11T single shank ripper assembly:

Distance between outer pin faces (assembled) = 847mm

Hole depth (each pin) = 63mm

Hole diameter = 36mm



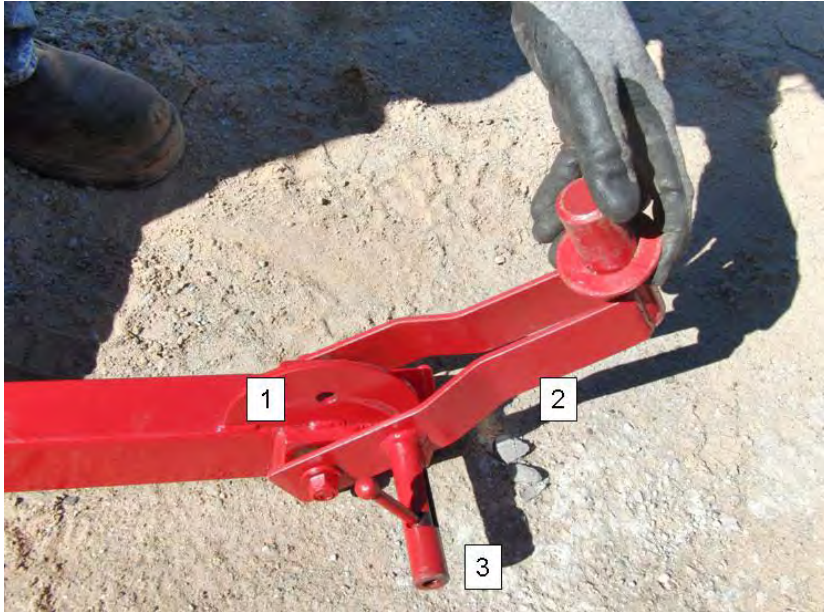
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Swivel End of Tool:

Key Design Requirements:

1. Pin rotate / locking plate
2. Rotating assembly
3. Spring loaded locking pin

Fixed End of Tool:

Key Design Requirements:

1. Manual locking pin



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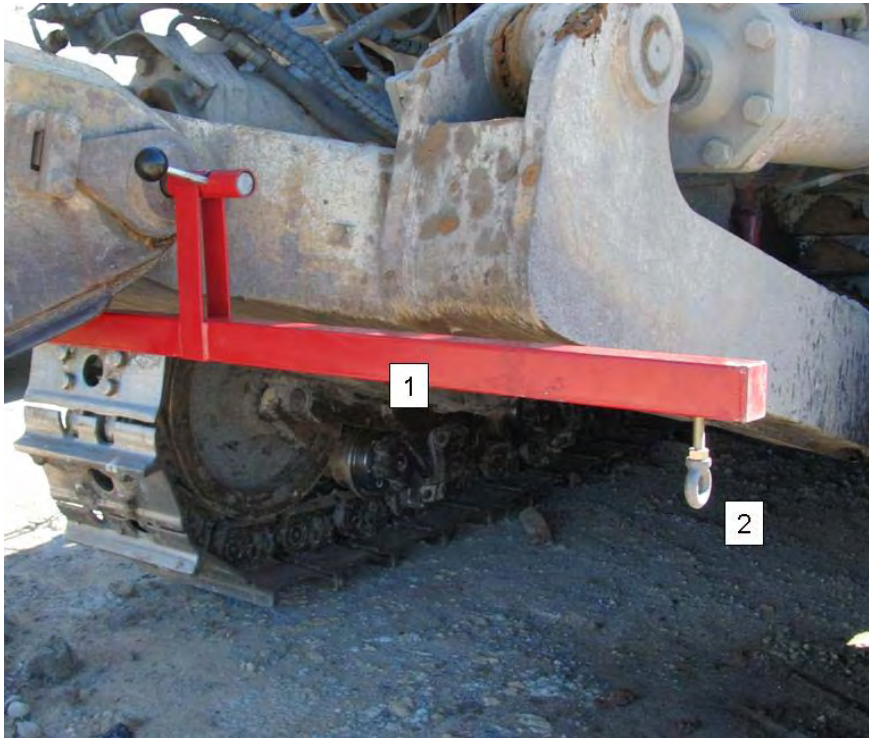
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Rail Assembly:

Key Design Requirements:

1. Lipped C Channel – Heavy Duty
2. Rail hook & roller



NOTE: For US users, any lifting tool must be MSHA certified for weight carrying capacity. In all countries, make sure you get lifting devices certified to adhere to local requirements.

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Shank Protector Ratchet Assembly:

Key Design Requirements:

1. Side mounted belt tensioner
2. Twin carabineers
3. Hooked ratchet strap ends

Ripper Protector Replacement Procedure:**Step 1.** Ripper set-up

Ripper pin puller to be used to extend out ripper shank to allow approximately 1.5 meters of shank to be extended below the ripper frame.

Step 2. Isolate Machine

Make sure that the machine is parked on a flat level surface, all implements are lowered, engine has been stopped and engine has been isolated (and tagged as appropriate).

Step 3. Prepare Lifting Frame

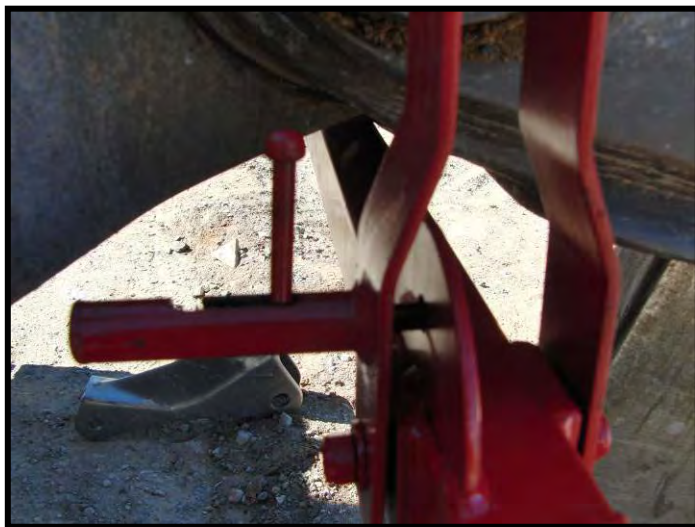
The spring loaded locking pin must be in the unlocked position and the Swivel pin opened out. The manual latch on the other end should also be in the unlatched position.

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**Step 4. Fit Swivel end of Lifting Frame to Ripper bed**

Insert the swivel end pin into the the ripper tilt pin. Lift up the lifting frame from opposite side (manual latch side) until the locking pin has clicked into the locked position. A distinct “click” should be heard when the pin is locked. Be sure that the swivel pin stays in position while lifting from the other end.



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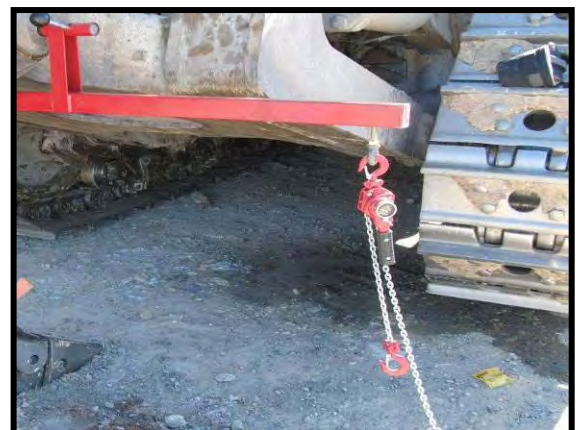
Step 5. Fit manual pin end of Lifting Frame to Ripper bed

Once the swivel pin end is locked into position, push the manual gun barrel style pin into position and lock it into place. The lifting frame should now be secure, test it by moving the frame side to side and back and forth to ensure it is safe.



Step 6. Attach ratchet chain block to lifting frame
Place the ratchet chain block onto the rail of the lifting frame. Make sure hook is secure and cannot fall off the rail or come lose.

Step 7. Ripper guard set-up
The ripper shin guard should be placed so the ripper side is facing down. This allows the ratchet strap to be attached in the correct position.



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Step 8. Fit ratchet strap to ripper guard

Position the ratchet strap, so when tightened, the lifting point will be just below the top of the shank. The correct location of the ratchet strap will make the guard positioning easier.

**Step 9.** Lift ripper guard.

Attach the chain and block hook onto the lifting rings of the ratchet strap. Ensure that it is attached securely and cannot fall. Lift up the guard and slide along the lifting frame rail until the guard is positioned above the ripper shank.



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Step 10. Lower ripper guard

Carefully lower the ripper guard onto the shank. Ensure that the retainers are in the correct position. When the shin guard is aligned, the retaining pins can be knocked into place using a soft face hammer.

**Step 11. Remove ratchet strap and chain.**

After the ripper shin guard is secure, release the tension from the chain block and un-hook the ratchet strap. Remove the chain block from the lifting frame rail.

Step 12. Unlock Swivel

Pull back the locking pin on the swivel end of the lifting frame. The pin should be moved around to the unlocked position.

Step 13. Remove lifting frame

Holding the frame from the manual pin end, slowly pull the frame away from the machine so the manual pin is clear of the machine. Once the manual pin is free, lower the bar slightly and push the bar so the swivel end releases from the machine.

3.0 Implementation Steps

1. Design and fabricate the fixture
Ensure fixture is certified to local standards as required.
2. Document the removal and installation procedure
Provide easy to follow instructions, use clear pictures and detailed instructions. Safety warnings and helpful tips should also be included.

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3. Train technicians
Teach technicians how to set up the machine properly including how to position the ripper, how to isolate and immobilise the machine.
4. Follow up and document changes or improvements

4.0 Benefits

- Improved safety – not lifting heavy objects
– less chance of injuring fingers
- Reduced labour – only one person required for R&I
- Reduced time to complete job

5.0 Resources Required

- Costs to design and fabricate fixture
- Costs to certify fixture
- Ratchet strap to hold ripper guard – rated to suitable limit

6.0 Supporting Attachments / References

- Ripper Guard Installation Tool.wmv

7.0 Related Best Practices

8.0 Acknowledgements

This Best Practice was written by:

- o Name: Brent Mekis
- o Title: CGM Marketing Representative
- o Email: mekis_brent@cat.com
- o Phone: +61 03 9953 9121

- o Name: Hayden Jones
- o Title: Product Performance Engineer
- o Email: hjones@cat.com
- o Phone: +61 03 9953 9414

Special thanks in preparation of this Best Practice to Craig Glanville from Rio Tinto.

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